A Positive Result Is Not a Probability of Truth

Week 10 Tuesday — Probability, uncertainty, and evidence

COMSC 2043 • Discrete Structures and Critical Thinking

October 20, 2026

Today's destination

- Examine the claim "90% accurate means 90% likely true."
- Build a four-cell count table and calculate $P(C \mid +)$.
- Change the base rate and predict first.
- See what a simulation checks, and what it cannot establish.
- Draft a one-sentence conclusion for the Decision Gate.

AI Fluency: examine one sentence

A claim someone might paste into a chat

"The test is 90% accurate, so a positive result means you are 90% likely to have the condition."

- Two different probabilities hide inside "90%."
- Which one is the test's *sensitivity*? Which one would the claim need?
- Name the ambiguity before doing any arithmetic.

Vocabulary we will use

Sample space every outcome we agree is possible.

Event a set of outcomes, e.g. C = "has the condition," $+$ = "tests positive."

Complement "not C": everything in the sample space outside C.

Conditional $P(A \mid B)$ = share of B that is also A; the denominator changes.

Sources and rules for the screening example

- **Source:** a clinic screens 10,000 people. Prevalence is 1%. Sensitivity is 90%. Specificity is 95%.
- **Sensitivity:** $P(+ \mid C) = 0.90$. **Specificity:** $P(- \mid \text{not } C) = 0.95$.
- **Assumption:** these rates apply to this population.
- Name one reason that assumption could be unsafe.

Natural frequencies: the four-cell table

	Tests $+$	Tests $-$	Total
Has C (1%)	90	10	100
Not C (99%)	495	9,405	9,900
Total	585	9,415	10,000

- 90% of 100 is 90. 5% of 9,900 is 495.
- Check: $90 + 10 + 495 + 9405 = 10{,}000$.

Reading the table: two different conditionals

$P(+ \mid C)$ — **sensitivity**

Denominator: everyone with C (100).
$90/100 = 90\%$

$P(C \mid +)$ — **what a positive means**

Denominator: everyone who tested positive (585).
$90/585 = 2/13 \approx 15.4\%$

Not the same number

A positive result here is correct about 15 times in 100, not 90.

Check, then change the base rate

1. Check: cells total 10,000, and the positive column (585) is the denominator.
2. **Predict first:** prevalence rises from 1% to 20%. Does $P(C \mid +)$ go up, down, or stay put?
3. Then calculate with the same sensitivity and specificity.

	Tests $+$	Tests $-$	Total
Has C (20%)	1,800	200	2,000
Not C (80%)	400	7,600	8,000
Total	2,200	7,800	10,000

Base-rate neglect

- Same test: $P(C \mid +) \approx 15.4\%$ at 1% prevalence, $\approx 81.8\%$ at 20%.
- The test did not change. The population did.
- Ignoring how common the condition is, and quoting only the test's error rates, is **base-rate neglect**.
- A claim about a positive result needs the base rate in the Sources line.

Independence and the gambler's fallacy

- Two events are **independent** if knowing one does not change the probability of the other.
- Assumption for the exercise: a fair coin, flips independent.
- Five heads in a row has probability $(1/2)^5 = 1/32 = 0.03125$ *before* you flip.
- Given five heads already happened, the next flip is still $1/2$: the coin has no memory.

What a simulation checks

```
import random
random.seed(10)
n, pos, pos_c = 1_000_000, 0, 0
for _ in range(n):
    c = random.random() < 0.01
    p = random.random() < (0.90 if c else 0.05)
    if p:
        pos += 1
        pos_c += c
print(pos_c / pos) # 0.1521... (calculation: 0.1538...)
```

- Agreement with 2/13 checks that the *code* implements the stated rates.

What a simulation does *not* establish

- It runs the assumptions you typed in. If 1%, 90%, and 95% are wrong for real patients, the simulation is wrong in exactly the same way.
- Agreement between table and simulation says nothing about whether the population matches the assumptions.
- More runs shrink noise, not assumption error.

Boundary note

Calculation and simulation agree $\Rightarrow$ the code matches the model. They do **not** agree $\Rightarrow$ "the model matches the world."

The AI base-rate check

AI answer to examine

"A positive result is correct 90% of the time, because the test is 90% sensitive."

- Which conditional probability did it confuse? ($P(+ \mid C)$ for $P(C \mid +)$.)
- What is missing from its Sources line? (The base rate.)
- Independent check: the four-cell table gives $90/585 \approx 15.4\%$.

The shape of every answer this week

1. **Sources** — rates, counts, rules, or simulation input.
2. **Rules / Assumptions** — event definitions; independence, representativeness, randomness.
3. **Work** — count table, calculation, or reproducible simulation.
4. **Check** — totals, a second calculation, a changed base rate; say what it cannot validate.
5. **One-Sentence Summary** — no stronger than the evidence.

Example: "In this 10,000-person example with 1% prevalence, about 15 of every 100 positives have the condition; this depends on the stated rates holding for the population."

Decision Gate note, and Thursday

- **Decision Gate:** one individual submission this week, using the five headings above. It is not an extra essay or a copy of Show & Tell.
- Choose the screening example, a stated game or drawing process, or an instructor-approved bounded claim.
- Name where AI helped, and check the arithmetic and interpretation yourself.
- **Thursday:** Show & Tell defense with a calculation and an evidence table or simulation.

The due date and time are on the assignment in Canvas.

Exit trace and what comes next

Keep your trace: the 1% table, $P(C \mid +)$, the 20% prediction and result, your boundary note, and one bounded sentence.

- Next week: relations. Which pairs of things are related, and how do we classify the relation?